

PRODUCT REQUIREMENTS DOCUMENT

Multi-Tenant AI Lead Calling Platform

HubSpot → n8n → AI Voice → Qualification → Google Sheets + HubSpot

Document Control	Value
Version	1.0
Date	1 September 2026
Status	Build-ready baseline
Product Type	B2B multi-tenant lead qualification / calling platform
Primary Stack	Next.js + PostgreSQL + n8n + HubSpot + Google Sheets + Voice/AI provider
Target Market	Multi-client lead generation and sales qualification
Primary Outcome	Automatically qualify inbound leads by voice and route high-intent leads to humans

Design principle One reusable calling platform; client-specific behavior comes from configuration and tenant-scoped data, not duplicated workflows.

This PRD is intentionally written as an implementation specification rather than a marketing brief. It defines the MVP, target architecture, workflows, data model, non-functional requirements, acceptance criteria, rollout plan, and key risks.

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1. Executive Summary

The product is a multi-tenant AI lead-calling platform for a lead-generation operation that serves multiple clients across different business domains. New leads arrive from each client’s HubSpot account. The platform validates and queues the lead, starts a compliant outbound voice interaction, asks domain-specific qualification questions, interprets the conversation, assigns an intent/qualification result, records the call, updates the client’s HubSpot record, and appends a structured row to the client’s Google Sheet.

The system must support many clients without creating separate application code or duplicated n8n workflows. Client-specific differences—industry, campaign, qualification questions, call script, scoring rules, calling windows, Google Sheet destination, HubSpot credentials, notification rules, and sales-routing logic—must be represented as data/configuration in a tenant-aware control plane.

Core architecture Next.js control plane + PostgreSQL system of record + n8n orchestration + voice provider + LLM/AI analysis + HubSpot/Google integrations.

Primary business value

- Reduce manual first-touch calling effort and response time.
- Standardize qualification across campaigns and clients while preserving domain-specific questions.
- Surface high-intent leads to human sales teams quickly.
- Create a complete, auditable history of calls, outcomes, transcripts, callbacks, and CRM updates.
- Operate multiple clients from one secure platform with strict tenant isolation.

2. Problem Statement

Current lead-generation operations commonly lose value between lead creation and human follow-up. Leads may sit in CRM queues, be called inconsistently, receive different questions depending on the caller, or have outcomes recorded manually in spreadsheets. With multiple clients in different domains, duplicating automations per client creates maintenance overhead and increases the risk of cross-client data contamination.

Observed problem pattern

Problem	Impact	Required product response
New leads are not called immediately	Lower contact and conversion rates	Event-driven intake + queue
Manual qualification varies by caller	Inconsistent lead data	Structured qualification schema
Each client has different questions	Hardcoded workflows become brittle	Campaign configuration
Multiple HubSpot accounts	CRM updates are hard to centralize	Per-tenant integration credentials
Google Sheets used operationally	Manual logging and duplicate work	Automatic append + retry
Human sales teams need hot leads fast	Lost high-intent opportunities	Real-time routing + notifications
Voice/commercial calling has regulatory constraints	Compliance and reputational risk	Consent/DNC controls + provider/legal review

3. Product Vision & Goals

Vision

Provide a reusable “AI first-touch sales layer” that can qualify leads for many clients and industries without cloning business logic.

Goals

- G1 — Multi-tenant: onboard a new client largely through configuration.
- G2 — Speed: get eligible leads into the calling queue within 30 seconds of CRM ingestion under normal load.
- G3 — Qualification: produce a structured result for at least 95% of successfully connected calls.
- G4 — Traceability: every attempted call has a durable call record and outcome.
- G5 — Routing: hot/high-intent leads are routed to humans within 60 seconds of qualification.
- G6 — Operational visibility: operators can see queue, calls, failures, callbacks, and client-level KPIs.
- G7 — Safety: enforce tenant isolation, DNC suppression, credential segregation, and auditable configuration changes.

Non-goals for MVP

- Fully autonomous sales closing or price negotiation.
- Replacing a human salesperson for complex product/service advice.
- Universal calling compliance certification for every geography.
- Building a custom telecom carrier network.
- Perfect sentiment/emotion detection.
- Real-time BI across every external analytics platform.

4. Personas & Users

Persona	Needs	Permissions
Agency Admin	Onboard client accounts, manage integrations and credentials, monitor all tenants	Global
Campaign Manager (Agency)	Configure campaigns, scripts, qualification fields on the client's behalf; the client does not log in or self-configure	Assigned tenant(s)
Operations Manager (Agency)	Monitor queues, failures, callbacks, daily performance across the agency's book of clients; owns the human-review queue (Section 26.3)	Tenant / assigned clients
Client Sales Recipient	Receives hot-lead notifications (email/Slack/CRM task) to follow up on; does not log into the platform and has no configuration access	None (notification-only, outside the platform)
Analyst	Review outcomes, conversion metrics, exports	Read/reporting
System / n8n Worker	Execute workflows and integrations	Service identity only

5. Scope

MVP in scope

- Multi-tenant client and campaign management.
- HubSpot inbound lead trigger/ingestion.
- Lead deduplication and eligibility rules.
- Calling queue with retries and calling windows.
- AI voice conversation with configurable prompt/questions.
- Call status webhooks and transcript/result ingestion.
- LLM-based structured qualification.
- Google Sheets append logging.
- HubSpot record update.
- Hot lead/human follow-up routing.
- Callback scheduling.
- DNC suppression.
- Operational dashboard and audit log.

Later scope

- Human live transfer during a call.
- WhatsApp/SMS follow-up journeys.
- Advanced workforce management.
- A/B testing of scripts.
- Predictive conversion models trained on historical outcomes.
- Client self-service billing/usage management.

6. Functional Requirements

6.1 Tenant & Client Management

ID	Requirement	Priority	Acceptance
FR-001	Admin can create a client/tenant with status active/inactive.	P0	Tenant is created with unique ID and isolated data scope.
FR-002	Client can have multiple campaigns.	P0	Campaigns inherit tenant scope.
FR-003	Each tenant can have one or more HubSpot integrations.	P0	Credentials stored securely and selectable by campaign.
FR-004	Each campaign can map to a Google Sheet destination.	P0	Append target is configuration, not hardcoded.
FR-005	Client admins cannot access another tenant.	P0	Authorization tests prove denial across tenants.

6.2 Lead Intake

ID	Requirement	Priority	Acceptance
FR-010	Accept new-lead webhook events from HubSpot or n8n-compatible ingress.	P0	Valid event creates/updates lead record.
FR-011	Normalize phone numbers to	P0	Stored normalized number

	E.164 where possible.		used by calling service.
FR-012	Reject or quarantine leads without a callable phone number.	P0	Reason is recorded and no call is placed.
FR-013	Deduplicate by tenant + source record ID and secondary phone/email rules.	P0	Duplicate event does not create duplicate call.
FR-014	Apply consent/eligibility/DNC rules before queueing.	P0	Suppressed leads do not enter call queue.

6.3 Calling

ID	Requirement	Priority	Acceptance
FR-020	Queue eligible leads for outbound calling.	P0	Lead status becomes queued.
FR-021	Respect configurable campaign calling windows and timezone.	P0	Calls outside window are delayed.
FR-022	Limit concurrent calls per tenant/campaign/provider.	P0	Configured cap is enforced.
FR-023	Capture initiated/ringing/answered/completed/failed statuses where supported.	P0	Call timeline recorded.
FR-024	Support retry policies for no-answer/busy/transient failures.	P0	Retries follow configured schedule and stop after max attempts.
FR-025	Stop all future calling when DNC is set.	P0	Suppressed lead cannot be re-queued.

6.4 AI Qualification

ID	Requirement	Priority	Acceptance
FR-030	Campaign-specific opening and qualification prompts.	P0	AI uses active campaign configuration.
FR-031	AI must return strict structured result.	P0	Schema validation passes before persistence.
FR-032	Unknown answers may be marked unknown; AI must not invent values.	P0	Unsupported facts are null/unknown.
FR-033	Conversation can detect callback requests and DNC intent.	P0	Flags drive routing.
FR-034	Human escalation is supported as a distinct result.	P1	human_followup=true creates task/notification.
FR-035	Low-confidence or borderline qualification results are routed to a human review queue instead of being auto-committed to CRM/Sheets.	P0	Results below the confidence/consistency threshold are held in pending_review status and excluded from auto-sync until an operator confirms or corrects them.

6.5 Data & CRM

ID	Requirement	Priority	Acceptance
FR-040	Persist lead, campaign, call, transcript metadata, analysis, and routing events.	P0	Records retrievable by tenant.
FR-041	Append one row per call attempt to configured Google Sheet.	P0	Rows contain standardized columns.
FR-042	Update HubSpot lead/contact	P0	CRM reflects current outcome.

	properties with latest qualification outcome.		
FR-043	Create or update callback/follow-up task.	P1	Task includes call context.

7. End-to-End User & System Flows

7.1 Happy path

```

HubSpot Lead Event
-> Ingress
-> Validate + Normalize
-> Tenant/Campaign Resolution
-> Consent/DNC Check
-> Queue
-> Calling Worker
-> Voice Conversation
-> Provider Webhook
-> Transcript/Result
-> LLM Qualification
-> Schema Validation
-> PostgreSQL
-> Google Sheets Append
-> HubSpot Update
-> Route Hot Lead / Callback / Stop

```

7.2 No answer

```

Lead -> Call -> no_answer -> call_attempt++ -> retry policy -> requeue OR terminal
no_answer

```

7.3 Interested lead

```

Call -> intent=hot/interested -> score>=threshold -> create sales task -> notify assigned
user -> update HubSpot -> log event

```

7.4 DNC during conversation

```

Caller says "do not call me" -> AI sets do_not_call=true -> persist DNC -> stop retries -
> update CRM -> suppress future campaigns

```

8. Multi-Tenant Architecture

8.1 Logical components

Component	Responsibility
Next.js App	Control plane UI, auth, configuration, reporting, manual actions
PostgreSQL	System of record, queues/state, tenant configuration, audit events
n8n	Event orchestration, integrations, retries, async workflow execution
Voice Provider	PSTN outbound calling, call status, media/recording as supported
AI Voice/LLM Layer	Speech interaction and/or transcript analysis
HubSpot	Client CRM/source of lead

Google Sheets	Operational client-facing append log
Notification Layer	Email/Slack/WhatsApp/SMS as configured later

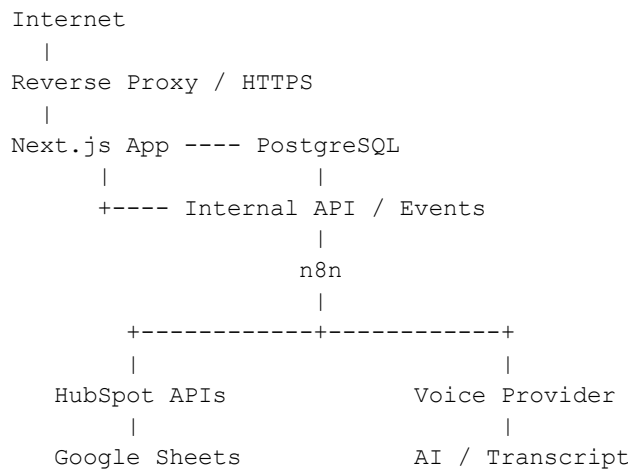
8.2 Tenant isolation

Every business record must carry `tenant_id`. Application authorization must derive tenant context from the authenticated session, not from user-supplied request fields. Service workflows must resolve `tenant_id` from trusted integration metadata or signed/internal references. Cross-tenant queries must be structurally prevented, not merely filtered in the UI.

Because the agency's own staff hold full access across every client's account, the primary isolation risk in this platform is an internal operator viewing, editing, or exporting the wrong client's leads — not an external attacker probing for tenant IDs. Role assignment (Section 4) must scope each Campaign Manager and Operations Manager to only their assigned client tenant(s) by default; access to a client outside that assignment requires an explicit, logged elevation rather than being available by default because the agency "has access to everything."

Rule No workflow or API endpoint may accept an arbitrary `tenant_id` from a public client and trust it as authorization.

8.3 Deployment topology



9. n8n Workflow Specifications

Workflow W01 — Lead Intake

Step	Node / Action	Output
1	Webhook / HubSpot trigger	Raw lead event
2	Normalize	Canonical contact payload
3	Resolve tenant/campaign	<code>tenant_id</code> , <code>campaign_id</code>
4	Check lead eligibility	eligible/suppressed
5	Idempotency check	new/duplicate
6	Persist lead	<code>lead_id</code>
7	Queue call	queue state
8	Acknowledge	event accepted

Workflow W02 — Calling Worker

Step	Node / Action	Output
1	Schedule/queue trigger	next eligible leads
2	Lock/claim lead	worker ownership
3	Check calling window	allowed/delayed
4	Mark calling	call session
5	Start provider call	provider call ID
6	Wait via webhook	call events
7	Persist final call	duration/outcome
8	Trigger qualification	analysis job

Workflow W03 — Qualification

Step	Node / Action	Output
1	Get transcript/result	conversation
2	Load campaign rules	schema + rubric
3	LLM analysis	JSON candidate
4	Validate schema	valid/invalid
5	Persist analysis	analysis_id
6	Calculate routing	hot/callback/stop
7	Google Sheets append	row logged
8	HubSpot update	CRM sync
9	Notify human	optional task/message

Workflow W04 — Retry / Callback

Trigger	Action
no_answer	increment attempt; schedule retry within campaign policy
busy	schedule short retry unless callback time exists
callback_requested	schedule exact callback time if available
provider_failure	retry transient failures with exponential backoff
DNC	terminate retries permanently
max_attempts_reached	terminal status; no more calls

n8n engineering rules

- Use production webhook URLs only for live integrations; test URLs only for development. n8n documents separate test and production webhook URLs and requires published workflows for production webhooks. [Ref. 1]
- Do not store call state only in n8n execution history; persist durable state in PostgreSQL.
- Use correlation IDs across HubSpot event → lead → call → analysis → CRM update.
- Design every outbound integration step to be idempotent or deduplicated.
- Record workflow version/config version with each call for auditability.

10. AI Voice Agent Specification

10.1 Agent responsibilities

- Identify the business/client context provided by the campaign configuration.
- Use a concise opening that truthfully explains the reason for the call.
- Ask only the qualification questions necessary for the campaign objective.
- Handle common conversational detours without inventing product facts.

- Detect positive/negative/uncertain intent.
- Detect callback requests, explicit DNC requests, wrong numbers, and human requests.
- End the call politely and provide the configured next step.

10.2 Agent constraints

- Never claim to be a human if it is an AI agent; disclose identity as required by policy/law/brand configuration.
- Never fabricate pricing, availability, eligibility, policy terms, or product details.
- Never override DNC or other suppression state.
- Never continue after a clear DNC request.
- Escalate to human when configured thresholds are met.

10.3 Conversation structure

Stage	Objective	Typical fields
Opening	Context + permission to continue	intro, availability_to_talk
Need discovery	Confirm ongoing need	still_interested, need
Qualification	Collect campaign-specific facts	budget, timeline, location, product
Intent check	Determine seriousness	intent, score
Next step	Callback / human / stop	callback_time, human_followup
Close	Confirm next step politely	next_action

10.4 Voice provider abstraction

Implement a provider adapter so the core platform is not coupled to a single telecom or AI vendor. Provider contract should expose: `create_call`, `get_call_status`, `handle_webhook`, `retrieve_recording_or_transcript`, `hangup`, and `provider_metadata`. This allows regional providers to be used when carrier/regulatory requirements make a global provider unsuitable.

India calling constraint Current provider guidance must be checked for India-specific outbound restrictions and commercial-call consent requirements. For example, Twilio's current India voice guidance says outbound calls to Indian non-Twilio numbers can only be made from international (non-Indian) numbers and recommends legal review; this makes provider selection a compliance/fit decision, not a default technology choice. [Ref. 7]

11. Lead Qualification & Scoring

11.1 Standard outcome taxonomy

Outcome	Definition	Typical action
hot	High purchase intent / ready for human follow-up	Immediate sales routing
interested	Genuine interest, not necessarily immediate	Sales task / nurture
warm	Some interest but weak commitment	Nurture / follow-up
follow_up	Explicit request to be contacted later	Schedule callback
not_interested	Explicitly declines	Stop campaign
wrong_number	Number not associated with lead	Stop / data correction
no_answer	No meaningful connection	Retry
busy	Connected or call rejected due to availability	Retry
do_not_call	Explicit no-call request	Permanent suppression

11.2 Score rubric

Signal	Example	Points
Current need confirmed	"I am actively looking."	+25
Short purchase timeline	0–3 months	+25
Budget/fit known	Provides actionable budget	+15
Specific product/service	Clear product selected	+10
Accepts human follow-up	Requests advisor	+15
Vague curiosity	"Just checking."	+5
Long-term/no plan	12+ months	0
Explicit rejection	Not interested	-100

The scoring rubric must be configurable per campaign. The default rubric is illustrative and must not override domain-specific qualification logic.

11.3 Routing thresholds

Condition	Action
score >= 75	Hot; human sales follow-up
50–74	Interested/warm; create follow-up
1–49	Nurture or classify based on intent
intent=do_not_call	Suppress regardless of score
intent=not_interested	Stop campaign unless user re-engages
callback_requested=true	Create callback regardless of score

12. Data Model

Core entities

Entity	Key fields
tenants	id, name, domain, status, timezone, created_at
users	id, tenant_id, role, email, status
integrations	id, tenant_id, type, credential_ref, status
campaigns	id, tenant_id, name, domain, script, timezone, active
qualification_rules	id, campaign_id, field_name, question, required, rubric
leads	id, tenant_id, campaign_id, hubspot_record_id, name, phone, email, source, status, dnc
consents	id, tenant_id, lead_id, basis, source, captured_at, captured_by, evidence_ref, status, revoked_at
call_attempts	id, lead_id, attempt_no, provider, provider_call_id, started_at, ended_at, status, duration, recording_ref
call_transcripts	id, call_id, transcript_ref, language, created_at
call_analyses	id, call_id, intent, score, qualification, structured_payload, model, prompt_version
callbacks	id, lead_id, requested_at, scheduled_for, status
routing_events	id, lead_id, action, assignee, status, created_at
audit_events	id, tenant_id, actor_type, actor_id, action, entity_type, entity_id, metadata, created_at

Indexes

- (tenant_id, status) on leads and calls.
- (tenant_id, hubspot_record_id) unique where source record is unique.
- (tenant_id, normalized_phone) for duplicate detection.

- (campaign_id, status, next_call_at) for queue workers.
- (call_id) on analyses and transcripts.
- (tenant_id, created_at) on audit and routing events.

13. Integrations

13.1 HubSpot

HubSpot is the primary CRM/source of lead context. Current HubSpot CRM contact APIs support updating individual contacts or batches, including using record IDs or unique properties. [Ref. 2]

Integration	Direction	Key fields
HubSpot Lead Event	HubSpot -> platform	record ID, name, phone, email, campaign/source
Contact Update	platform -> HubSpot	call status, intent, score, qualification, callback
Task/Owner	platform -> HubSpot	follow-up task, owner, due date
Custom properties	platform -> HubSpot	dnc, last_call, next_call, AI outcome

13.2 Google Sheets

Google Sheets is an operational reporting/logging destination. The Sheets API supports appending values to the next row of a table/range and requires spreadsheet ID, range, and an input option. [Ref. 3]

- Prefer server-side OAuth/service-account style integration according to the chosen Google Workspace security model.
- Store destination spreadsheet ID and tab/range per campaign or tenant.
- Append after call completion; on retry, use call_id as idempotency key in the application before writing.
- Treat PostgreSQL as source of truth.

13.3 Voice / AI provider

Use an adapter pattern. Provider-specific implementation is responsible for telephone connectivity, call control, status callbacks, transcription/media where offered, and any provider-specific authentication.

14. Web Application / Admin UX

14.1 Navigation

- Dashboard
- Clients
- Campaigns
- Leads
- Calls
- Callbacks
- Analytics
- Integrations
- Settings
- Audit Log

14.2 Dashboard

Widget	Description
Leads today	New leads by client/campaign
Calls today	Attempted, connected, completed
Connection rate	Connected / attempted
Intent mix	Hot / interested / warm / not interested / no answer
Hot leads	Unassigned high-score leads requiring action
Callbacks due	Today/overdue callbacks
Failures	Provider, integration, workflow errors
Client health	Tenant-level SLA and queue backlog

14.3 Client onboarding wizard (agency-internal)

The client does not use this platform, log in, or configure anything directly. The agency holds full access to the client's HubSpot, Google Workspace, and (where applicable) voice provider accounts, and a Campaign Manager runs this wizard on the client's behalf, typically based on a kickoff call, brief, or brand/script materials supplied by the client outside the platform.

30. Create tenant and timezone for the client account.
31. Connect the client's HubSpot using agency-managed credentials.
32. Connect the client's Google Sheets destination (agency-managed service account or delegated access).
33. Add campaign and business description based on client-supplied brief.
34. Define call script/opening.
35. Define qualification questions and fields.
36. Configure scoring thresholds.
37. Configure calling hours/retries.
38. Configure human routing to the client's designated sales recipient(s) (Section 4) — notification channel only, no platform access granted.
39. Record the consent basis for this client's lead list (Section 26.1) before any test or live calling.
40. Run a test call in sandbox/staging mode.
41. Activate campaign.

14.4 Lead detail page

- Contact identity and source.
- Latest intent, score, qualification.
- Full call history.
- Call transcript/summary (according to access policy).
- Recording link where permitted.
- Next callback.
- CRM sync status.
- Audit timeline.
- Manual actions: suppress, retry, reassign, schedule callback.

15. Google Sheets Output

Call Log columns

Column	Example
--------	---------

call_id	C-20260901-000123
client_id	TEN-001
campaign	Noida 2BHK
lead_id	LEAD-12345
hubspot_record_id	123456789
name	Rahul Sharma
phone	+919876543210
call_date	2026-09-01
duration_sec	154
call_status	completed
intent	hot
score	82
qualification	qualified
timeline	0-3 months
budget	50-75 lakh
location	Noida
callback_requested	false
human_followup	true
dnc	false
summary	Actively searching; requests advisor
recording_ref	provider://...
sync_status	success

16. Notifications & Human Handoff

Hot lead notification

🔥 HOT LEAD
 Client: ABC Real Estate
 Lead: Rahul Sharma
 Score: 82
 Intent: Hot
 Timeline: 0-3 months
 Need: 2BHK in Noida
 Next step: Sales advisor requested
 Call: 2m 34s
 HubSpot: <record link>

Routing rules

- Assign by client, campaign, geography, product, or round-robin.
- Avoid duplicate assignment when multiple workflows receive the same result.
- Track acknowledgment/ownership where supported.
- Escalate stale hot leads after configurable SLA (e.g., 10 minutes).

17. Security, Privacy & Compliance

17.1 Security requirements

- HTTPS everywhere; secure cookies/session handling.
- Encrypt integration secrets at rest; store only secret references in app/database.
- Use least-privilege service accounts.
- Separate production/staging credentials.

- Do not expose transcripts, recordings, or phone numbers in public URLs.
- Audit configuration changes, manual suppression, routing changes, and data exports.
- Apply tenant-scoped authorization server-side.
- Implement rate limits on public webhooks and admin APIs.

17.2 Data retention

Data	Default policy proposal	Notes
Lead data	Until client-defined retention period	Client contract may override
Call metadata	12 months	Longer only if justified
Transcript	90 days	Configurable
Recording	30–90 days	Configurable, consent/legal review
Audit log	24 months	Security/compliance

Retention periods are product defaults for design purposes, not legal advice. Final retention must be aligned with customer contracts, applicable privacy law, telecom rules, sector rules, and provider terms.

17.3 India commercial calling

For India, the platform must include a compliance gate before live outbound commercial calling. TRAI's current sender guidance states that commercial communication may only take place using registered headers and distinguishes promotional versus service communications; promotional voice calls are described as commercial calls where explicit consent has not been obtained. [Ref. 4] TRAI's current consolidated regulations page lists the Telecom Commercial Communications Customer Preference Regulations, 2018 with a consolidated release dated 21 May 2026, and a March 2026 consultation described additional proposed amendments related to UCC. [Refs. 5–6]

Because the agency operates the calling infrastructure end-to-end and the client has no direct involvement in placing calls, the agency — not the client — is generally the party actually initiating outbound commercial communication and is the one whose sender/telemarketer registration, header registration, and consent-verification obligations are most directly engaged. The client's role is limited to supplying the lead list and its claimed consent basis; the agency is responsible for confirming that basis is adequate before dialing, and for its own registration status with the relevant telecom authority, independent of whatever the client has or hasn't registered on their own account. This should be confirmed with qualified telecom/legal counsel before launch, as the correct registration path can depend on the specific operating structure (e.g., whether the agency dials as itself, as a registered principal entity on the client's behalf, or the client is the registered principal and the agency operates strictly as its technical processor).

Required product behavior Do not activate India outbound campaigns until a telecom/compliance review confirms the agency's own sender/telemarketer registration and calling category, the consent basis and evidence supplied for the client's lead list, provider arrangement, DNC handling, recording notices, and retention requirements. A client's verbal assurance that "leads are opted in" is not sufficient evidence on its own — see Section 26.1.

17.4 DNC / suppression

- Global tenant DNC list takes precedence over campaign eligibility.
- Campaign suppression can be narrower; global DNC cannot be overridden by campaign configuration.
- Manual DNC must immediately invalidate queued calls.
- Voice-detected DNC must terminate future retries after the current call is processed.

18. Reliability, Observability & Operations

18.1 Idempotency

- HubSpot event idempotency key = tenant_id + source_event_id/record_id + event type.
- Call result idempotency key = provider + provider_call_id.
- Sheet write dedupe key = call_id.
- CRM update can be retried safely by setting the same current-state properties.

18.2 Failure classes

Failure	Retry?	Fallback
Temporary HTTP 5xx	Yes	Exponential backoff
Auth failure	No	Alert + integration disabled
Invalid lead phone	No	Quarantine lead
Provider busy/rate limit	Yes	Queue later
LLM schema failure	Yes, limited	Fallback to manual review
Google Sheets outage	Yes	Store pending sync in DB
HubSpot outage	Yes	Pending CRM sync
Webhook signature invalid	No	Reject + security log

18.3 Observability

- Central structured logs with correlation_id, tenant_id, workflow_id, lead_id, call_id.
- Metrics: queue depth, call latency, connect rate, analysis success rate, integration error rate.
- Alerts: queue backlog, provider failure spike, webhook failures, hot-lead SLA breaches, sync backlog.
- Dead-letter/pending state for records requiring manual replay.

19. Non-Functional Requirements

Category	Requirement	Target
Availability	Core application and workflows	99.5% monthly MVP target
Ingestion latency	HubSpot event -> queued	p95 < 30 sec
Routing latency	Call result -> human alert	p95 < 60 sec
API latency	Normal authenticated APIs	p95 < 500 ms excluding third-party calls
Scalability	Concurrent calls	Configurable; initial 25 concurrent system-wide
Durability	Call records	No silent loss; persisted before workflow completion
Security	Tenant isolation	100% authorization test coverage on tenant boundaries
Recovery	RPO	<= 15 min
Recovery	RTO	<= 2 hr
Auditability	Admin/config changes	100% logged

20. API / Event Contracts

Lead intake event

```
{
  "event_id": "evt_123",
  "tenant_ref": "TEN-001",
```

```
"source": "hubspot",
"record_id": "123456789",
"campaign_ref": "CAMP-001",
"contact": {
  "name": "Rahul Sharma",
  "phone": "+919876543210",
  "email": "rahul@example.com"
}
}
```

Call result event

```
{
  "call_id": "CALL-001",
  "provider_call_id": "CA123",
  "lead_id": "LEAD-001",
  "status": "completed",
  "duration_sec": 154,
  "transcript_ref": "tr_001",
  "recording_ref": "rec_001"
}
```

Qualification result

```
{
  "intent": "hot",
  "score": 82,
  "qualification": "qualified",
  "timeline": "0-3_months",
  "budget": "50-75_lakh",
  "location": "Noida",
  "product_interest": "2BHK",
  "callback_requested": false,
  "human_followup": true,
  "do_not_call": false,
  "reason": "Actively searching and requested advisor"
}
```

21. Analytics & KPIs

Metric	Definition	Purpose
Lead-to-call latency	first_call_started - lead_created	Operational speed
Connect rate	connected_calls / attempts	Reachability
Qualification completion	calls_with_structured_result / connected_calls	AI reliability
Hot rate	hot_leads / connected_calls	Lead quality
Human follow-up SLA	sales_ack - hot_result	Sales responsiveness
Callback completion	completed_callbacks / scheduled_callbacks	Follow-up quality
CRM sync success	successful_updates / update_attempts	Integration health
Cost per qualified lead	voice+AI cost / qualified leads	Unit economics
Lead conversion	won/custom outcome / leads	Business value

Client-level reporting

- Filter by tenant, campaign, source, date, geography, product.
- Compare campaigns on connect rate, intent mix, qualification rate and conversion.

- Separate AI performance metrics from commercial outcome metrics.

22. Implementation Plan

Phase 0 — Foundations

- Repository, environments, CI/CD.
- PostgreSQL schema and migrations.
- Auth/RBAC and tenant middleware.
- Secrets management.
- Base Next.js admin shell.

Phase 1 — Single-client vertical slice

- One HubSpot integration.
- One campaign.
- One voice provider adapter.
- One n8n intake workflow.
- One calling workflow.
- One qualification workflow.
- Google Sheets + HubSpot updates.
- End-to-end test lead.

Phase 2 — Multi-tenant

- Tenant-scoped config.
- Per-tenant credentials.
- Multiple campaigns.
- Per-campaign prompts/questions/scoring.
- Tenant dashboards.
- Audit logs.

Phase 3 — Production hardening

- Queue locking/concurrency controls.
- Retries and dead-letter handling.
- Observability and alerts.
- Load testing.
- Security testing.
- Compliance review and production calling approval.

Phase 4 — Advanced sales orchestration

- Human transfer.
- Callbacks.
- Multi-channel follow-up.
- Advanced routing.
- A/B testing.

23. Testing & Acceptance Criteria

23.1 Unit tests

- Tenant authorization.
- Phone normalization.
- Deduplication.
- Eligibility/DNC logic.
- Scoring rubric.
- Routing thresholds.
- Webhook signature verification.
- Schema validation.

23.2 Integration tests

- HubSpot event -> platform lead.
- Platform -> HubSpot property update.
- Call provider -> n8n webhook.
- n8n -> PostgreSQL.
- PostgreSQL -> Google Sheets append.
- Retry behavior on 429/5xx.

23.3 End-to-end acceptance scenarios

Scenario	Expected result
New eligible lead	Queued and called automatically
Duplicate lead event	No duplicate call
No phone	Quarantined; no call
Lead says interested	Structured interested/hot result persisted
Lead requests callback	Callback created at requested time
Lead says do not call	DNC stored; all future calls blocked
Provider timeout	Retry/failure handled without data loss
Sheets unavailable	Pending sync stored and replayed
HubSpot unavailable	CRM update queued for retry
Cross-tenant URL manipulation	403/404; no data disclosure
LLM returns invalid JSON	Validation fails; controlled retry/manual review

Definition of Done for MVP

- A real lead can flow from HubSpot to completed qualification without manual intervention.
- Every call has a durable call record.
- Google Sheet and HubSpot are updated from the same canonical result.
- A hot lead is visible to a human operator promptly.
- DNC behavior works across retries and campaigns.
- Tenant isolation passes security tests.
- Production runbook and rollback procedure exist.

24. Risks & Mitigations

Risk	Severity	Mitigation
------	----------	------------

Telecom/regulatory non-compliance	Critical	Legal review + compliant provider + consent/DNC controls
Poor call quality / STT errors	High	Provider testing, language evaluation, short prompts, fallback
Hallucinated qualification	High	Strict schema, allowed values, grounded campaign context
Cross-tenant data leakage	Critical	Tenant middleware, DB scope, service identity isolation
Third-party outages	High	Retries, pending sync, provider abstraction
Google Sheet used as queue	Medium	PostgreSQL system of record
Workflow duplication in n8n	Medium	Use DB state + idempotency keys
High cost per qualified lead	High	Track unit economics; optimize prompts and concurrency
Client-specific requirements explode complexity	High	Configuration model + explicit extension points

25. Future Roadmap

Feature	Value
Human live transfer	Fast escalation of hot leads
AI voice persona/config presets	Faster campaign onboarding
Multilingual conversations	Broader regional reach
WhatsApp follow-up	Multi-channel conversion
Lead reactivation campaigns	Monetize older CRM leads subject to consent
A/B testing	Optimize scripts
Predictive lead scoring	Improve prioritization
Client self-service portal	Reduce operations workload
Usage billing	Turn platform into SaaS
Conversation QA	Monitor agent quality and policy adherence

26. Consent, PII/Security & Human-Review Additions

This section closes three gaps identified during PRD review: consent is made a first-class, provable record rather than an implied state; PII protection is specified at the field level rather than assumed under general "encrypt secrets" language; and a human-review safety net is added for low-confidence AI qualification results. These additions are prerequisites for the India compliance gate in Section 17.3 and the Definition of Done in Section 23.

26.1 Consent data model

Consent must be captured as a durable, queryable record — not inferred from the fact that a lead exists in the queue. This matters more, not less, under a fully agency-operated model: the agency did not run the client's original opt-in funnel and has no independent way to verify a lead's consent beyond what the client supplied. The new consents entity captures that supplied basis explicitly, so the agency has its own evidentiary trail rather than an unverified assumption inherited from the client at intake.

Field	Purpose	Example
basis	Legal basis for contact	opt_in_form, existing_customer, service_call
source	Where consent was obtained	landing_page_form, hubspot_import, ivr_confirmation
evidence_ref	Pointer to durable proof	form submission ID, recorded IVR confirmation

status / revoked_at	Whether consent is still active	active, withdrawn, expired
---------------------	---------------------------------	----------------------------

Rule A lead may not be queued for calling unless an active consents record exists for it, or the campaign is explicitly flagged as service-call/existing-relationship by a Client Admin. The consent basis is written to the call_attempts record at call time so each call can be individually justified later.

26.2 PII & security addendum

Section 17.1 covers secrets and transport security. This addendum covers the lead/call PII itself, which is the more sensitive and higher-volume data class in this system.

Data class	Protection requirement	Rationale
leads.phone, leads.email	Column-level encryption at rest; masked in list views, unmasked only on explicit detail-view action	Direct identifiers across every tenant's real customers
call_transcripts, call_attempts.recording_ref	Access gated by role + assigned tenant; every read logged to audit_events	Recorded voice content is high-sensitivity and often independently regulated
call_analyses.structured_payload	Redact free-text fields on export; retain structured fields only past the transcript retention window	Summaries can still carry sensitive personal detail even after the transcript is deleted
Google Sheets export rows	Sheet access restricted to the owning client's Workspace principals only; no cross-tenant sharing links	Sheets sits outside the platform's own access-control layer once data leaves PostgreSQL

- Encrypt PII columns at the database or application layer, not just the disk/volume level, so a database dump alone does not expose raw phone numbers or emails.
- Data-subject deletion requests must cascade across leads, call_transcripts, call_analyses, and the corresponding Google Sheet row/HubSpot property, not just the primary lead record.
- Recording/transcript access by a Client Admin from a different tenant must be structurally impossible, not just hidden in the UI (same rule as Section 8.2, applied to media assets specifically).

26.3 Human-in-the-loop review queue

FR-035 (Section 6.4) requires that low-confidence or borderline AI qualification results are not auto-committed. This section defines what triggers that hold and what happens next.

Trigger	Behavior
LLM confidence below campaign threshold	call_analyses.qualification = pending_review; excluded from Sheets append and HubSpot update until resolved
Required qualification fields left unknown/null	Routed to review queue rather than defaulted to a score
Score near the hot/interested boundary (configurable band, e.g. ± 5)	Flagged for review; routing action held pending confirmation
Operator confirms or corrects	Final result released to Sheets/HubSpot/routing; correction logged to audit_events with actor_type=human
Review queue backlog exceeds SLA	Alert per Section 18.3; hot-intent items in the backlog escalate first

Why this matters Without this queue, a hallucinated "not interested" or a low-confidence "hot" silently determines a real lead's fate with no human ever seeing it. Because the client does not use the platform and never sees a result before it lands in their CRM, the agency's Operations Manager (Section 4) is the only checkpoint between an AI qualification call and the client's sales pipeline — there is no client-side review to catch a bad result downstream. The review queue is the safety net between "the model returned valid JSON" and "the result is actually trustworthy enough to act on."

Appendix A — Example Campaign Prompt

You are the AI lead qualification agent for {{client_name}}.

Business context:

{{business_description}}

Campaign:

{{campaign_name}}

Product/service:

{{product}}

Goal:

Determine whether the lead has genuine interest and whether a human sales follow-up is warranted.

Required questions:

{{questions}}

Rules:

- Be concise and polite.
- Do not invent facts.
- Ask one question at a time.
- Respect an explicit do-not-call request immediately.
- If the lead asks for a human, set human_followup=true.
- Return only structured data after the call is complete.

Appendix B — Example n8n Configuration Objects

```
{
  "tenant_id": "TEN-001",
  "campaign_id": "CAMP-001",
  "calling": {
    "timezone": "Asia/Kolkata",
    "window_start": "09:30",
    "window_end": "18:30",
    "max_attempts": 3,
    "retry_minutes": [15, 120, 1440]
  },
  "routing": {
    "hot_threshold": 75,
    "notify_channel": "sales_queue"
  }
}
```

Appendix C — References

The following sources were checked while preparing this PRD. They are included to ground the integration/compliance assumptions; final implementation should re-check vendor and regulatory documentation before production launch.

n8n Webhook documentation — <https://docs.n8n.io/integrations/builtin/core-nodes/n8n-nodes-base.webhook/>

n8n webhooks can receive external events and start workflows; production webhooks require published workflows. [Ref. 1]

HubSpot CRM contacts API — <https://developers.hubspot.com/docs/api-reference/latest/crm/objects/contacts/guide>

Current HubSpot docs describe individual and batch contact updates. [Ref. 2]

Google Sheets append API —

<https://developers.google.com/workspace/sheets/api/reference/rest/v4/spreadsheets.values/append>

Current API supports appending values to a detected table/range. [Ref. 3]

TRAI — Advice to Senders — <https://www.trai.gov.in/advice-to-senders>

Current TRAI guidance on commercial communications, registered headers, promotional voice calls, and service calls; updated 28 Aug 2026. [Ref. 4]

TRAI — Telecom regulations — <https://www.trai.gov.in/release-publication/consolidated-regulations/telecom>

Current consolidated telecom regulations page, including TCCCPR 2018 release dated 21 May 2026.

PIB — TRAI draft amendment consultation —

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2239885&lang=1®=1>

13 Mar 2026 announcement regarding draft third amendment consultation on commercial communications/UCC.

Twilio India voice guidelines — <https://www.twilio.com/en-us/guidelines/in/voice>

Current Twilio India voice guidance includes outbound restrictions and consent/legal review recommendations. [Ref. 7]

Implementation Decision Summary

Decision	Choice
Product model	Multi-tenant platform
System of record	PostgreSQL
Workflow engine	n8n
CRM	HubSpot per client/tenant
Operational log	Google Sheets per client/campaign
Frontend	Next.js
Voice layer	Provider abstraction; select India-capable/compliant provider
AI	Voice agent + structured LLM analysis
Tenant key	tenant_id on every business entity
Lead queue	Database-backed state + n8n workers
Primary MVP	Inbound HubSpot lead -> AI call -> intent -> Sheet + HubSpot -> sales routing

Build order Start with one client and one campaign as a vertical slice, then generalize into configuration-driven multi-tenancy before onboarding the second client. This is the fastest way to validate the core call/qualification loop without baking client-specific assumptions into the platform.